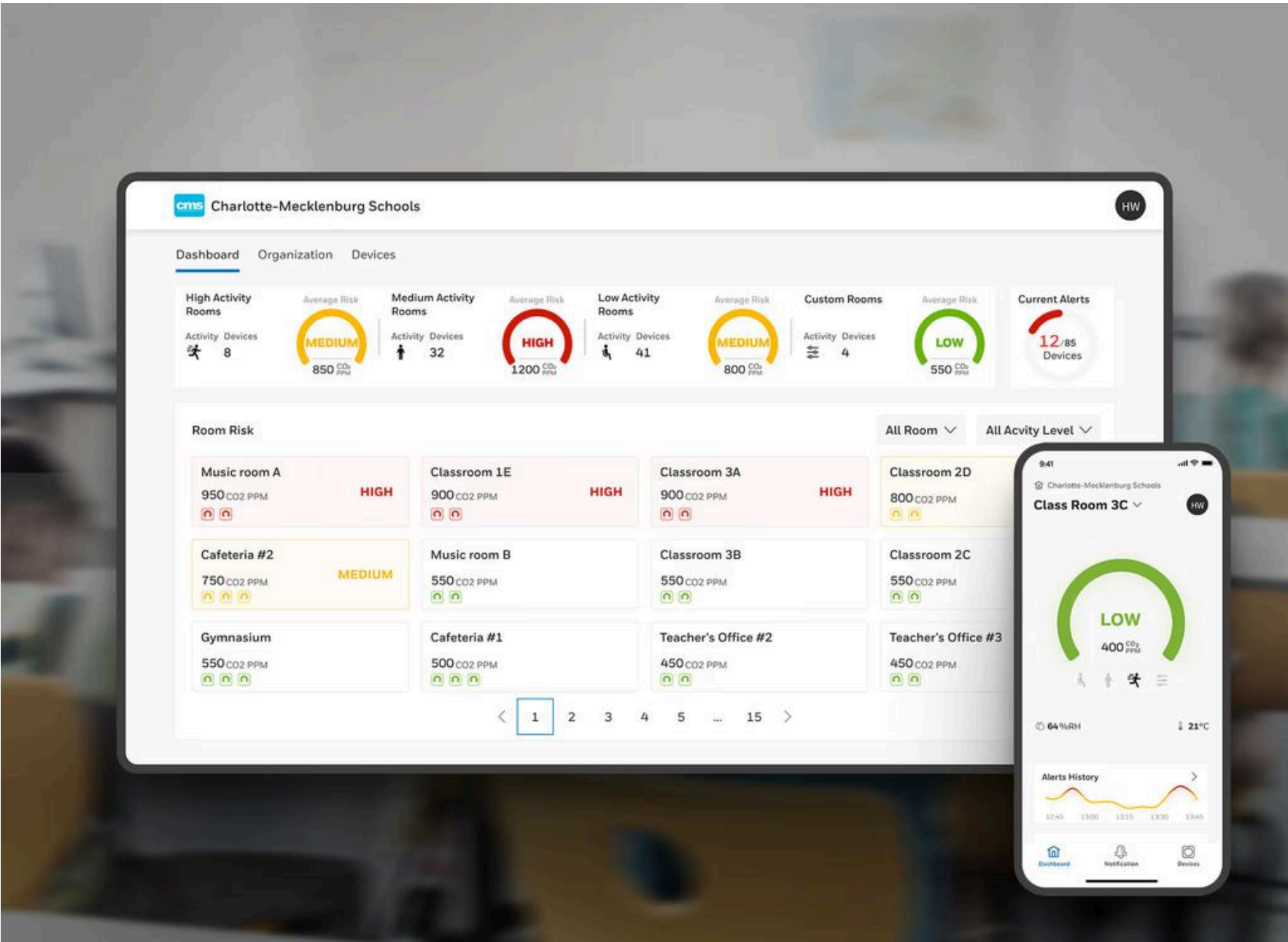
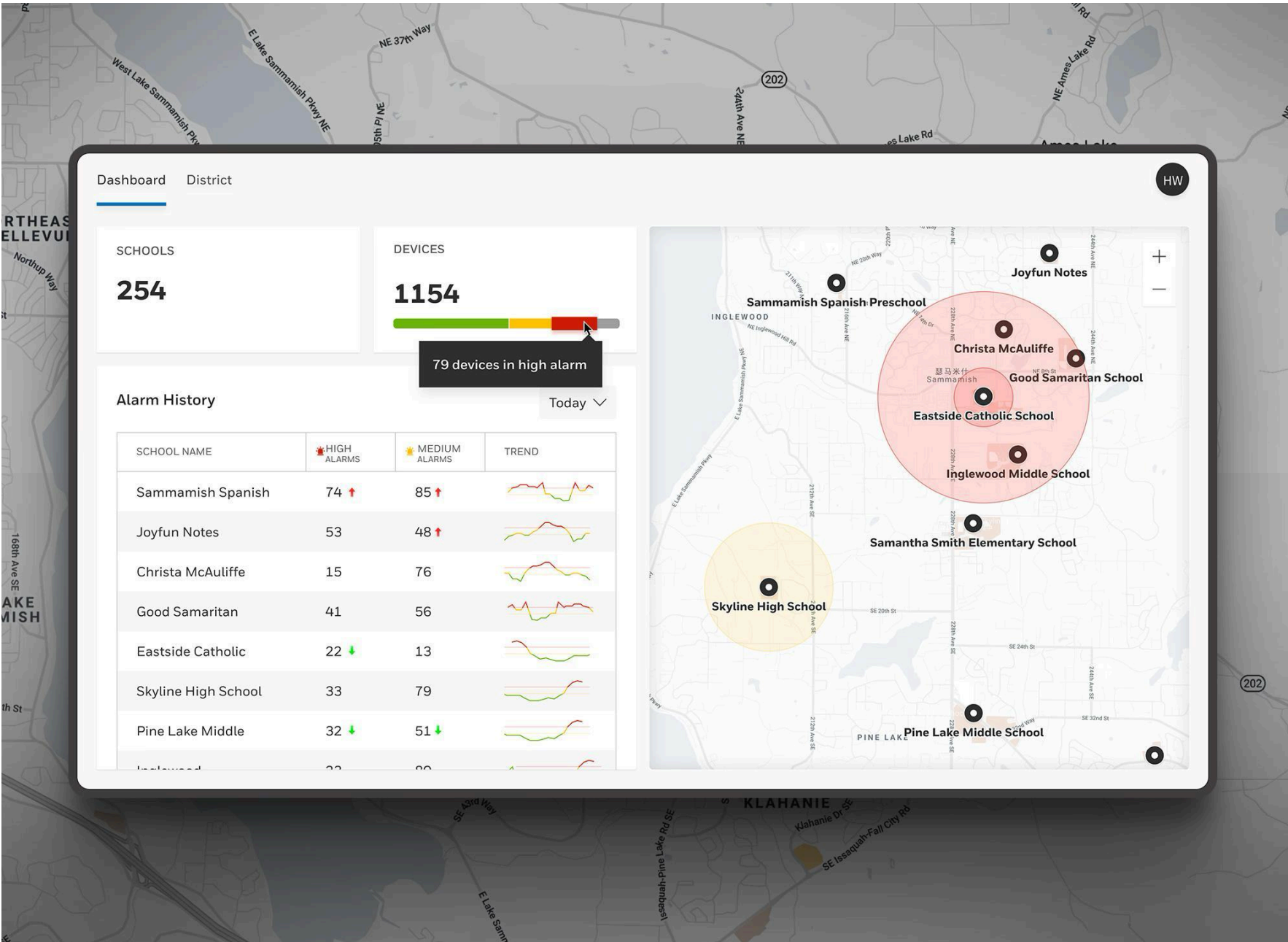


Honeywell's Tranmission Risk Air Monitor

From Data to Decisions: Designing for Breathable Air



I led the design of the Honeywell Transmission Risk Air Monitor experience, collaborating closely with a colleague to make air quality management intuitive for schools and public spaces. The challenge was to translate complex algorithmic risk assessments into clear, actionable guidance for teachers and administrators managing busy classrooms and schools. We streamlined mobile and web workflows, onboarding, and real-time alert dashboards, transforming data into actionable insights that support healthier, safer environment. The award-winning user interface empowered educators to create safer learning environments without becoming air quality experts.



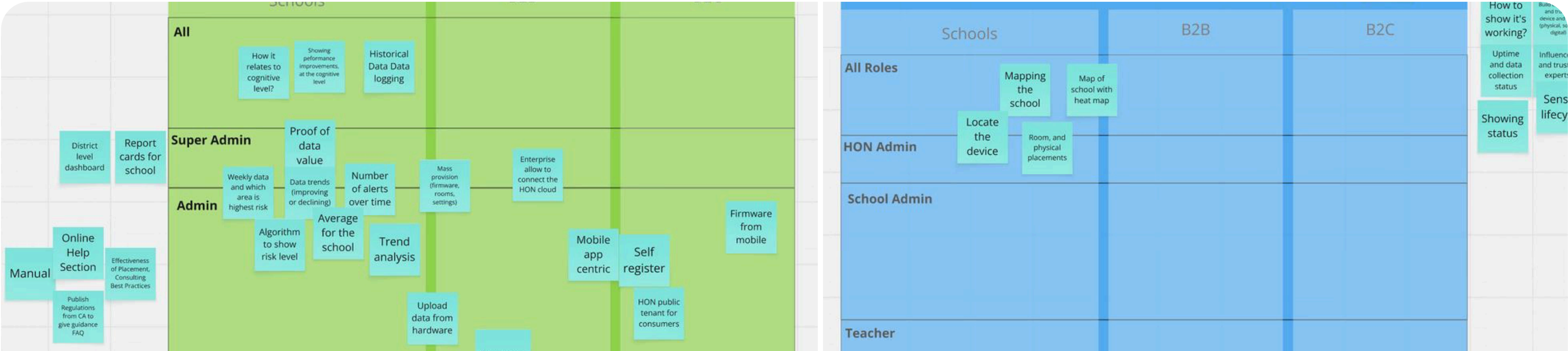


The Challenge

As awareness of indoor air quality grew, schools and public spaces increasingly needed simple, real-time indicators to identify when ventilation was inadequate. traditional HVAC systems typically operate on fixed schedules rather than adapting continuously to actual occupancy or CO₂ buildup, creating potential health and safety gaps.

Key design challenges included:

- Non-technical usability** — Users had to immediately understand device indicators without any training
- Intuitive setup** — Educators and administrators had to be able to complete device configuration independently
- Scalable software** — Mobile and desktop interfaces had to work seamlessly at the district, school, and room levels.
- System integration** — The solution required building system connectivity (HVAC, BMS) for automated response capabilities



Discovery

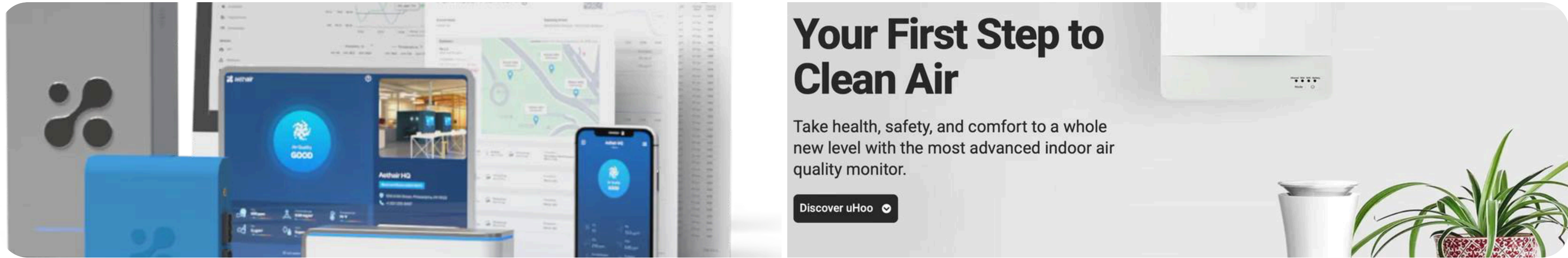
As UX Lead with a research and design role, I synthesized the insights from interviews, desktop research, competitive analysis and accessibility evaluations into themes to help define key features and priorities.

User research included interviews with facility managers, teachers, and school administrators to understand their expectations and constraints. How did they prefer to visualize and interpret air quality data? How would they realistically respond to alerts while managing noisy classrooms? What was the their comfort level with device setup processes involving WiFi and Bluetooth?

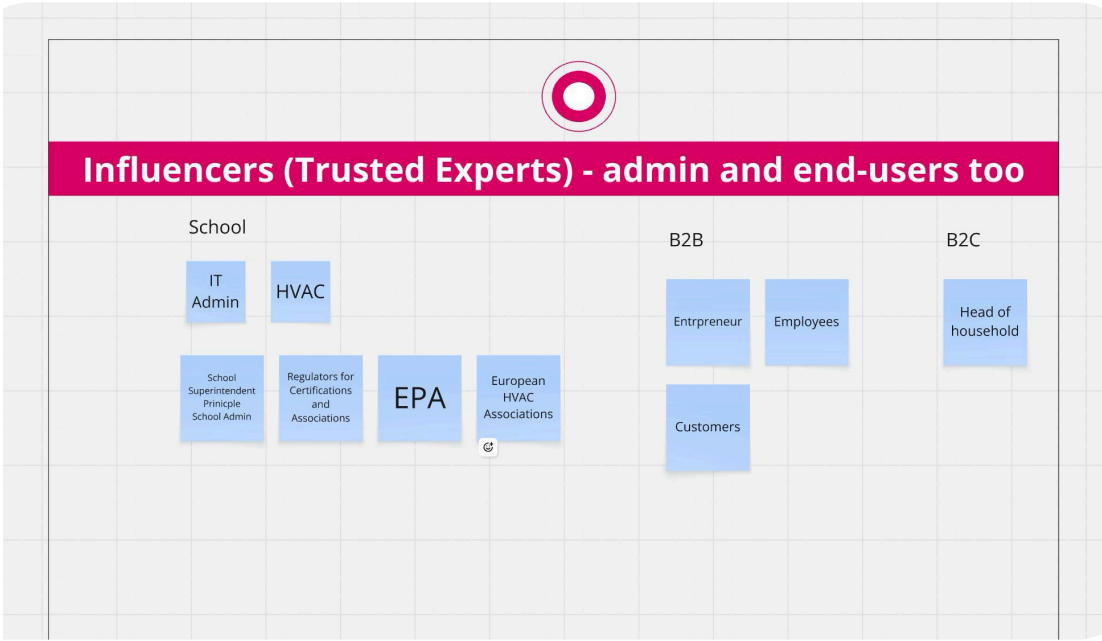
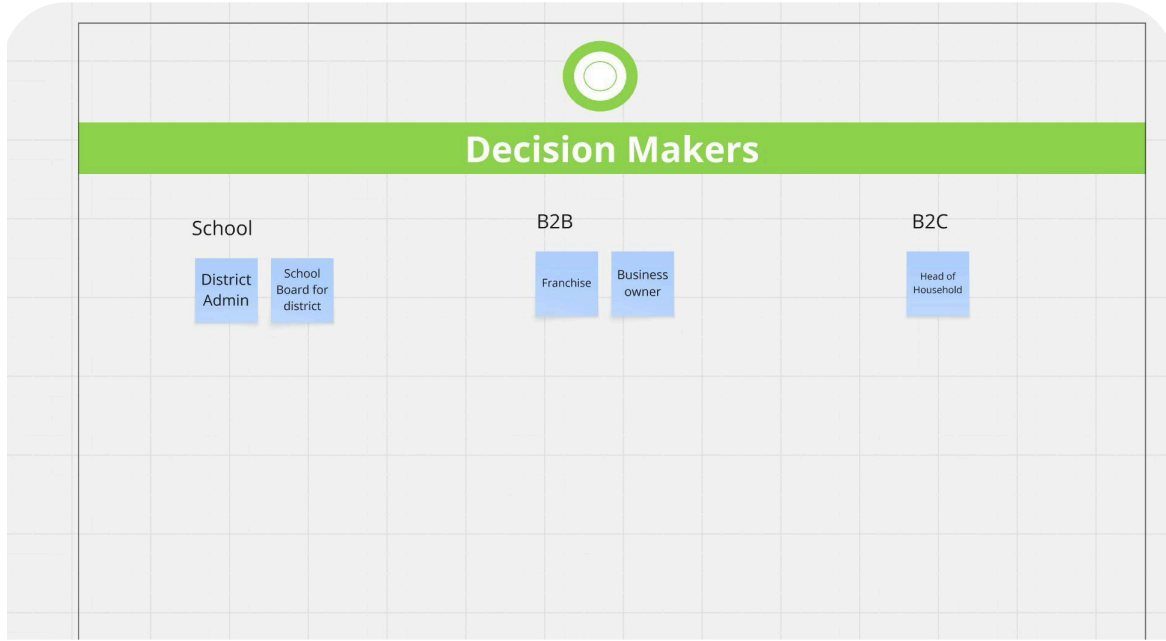
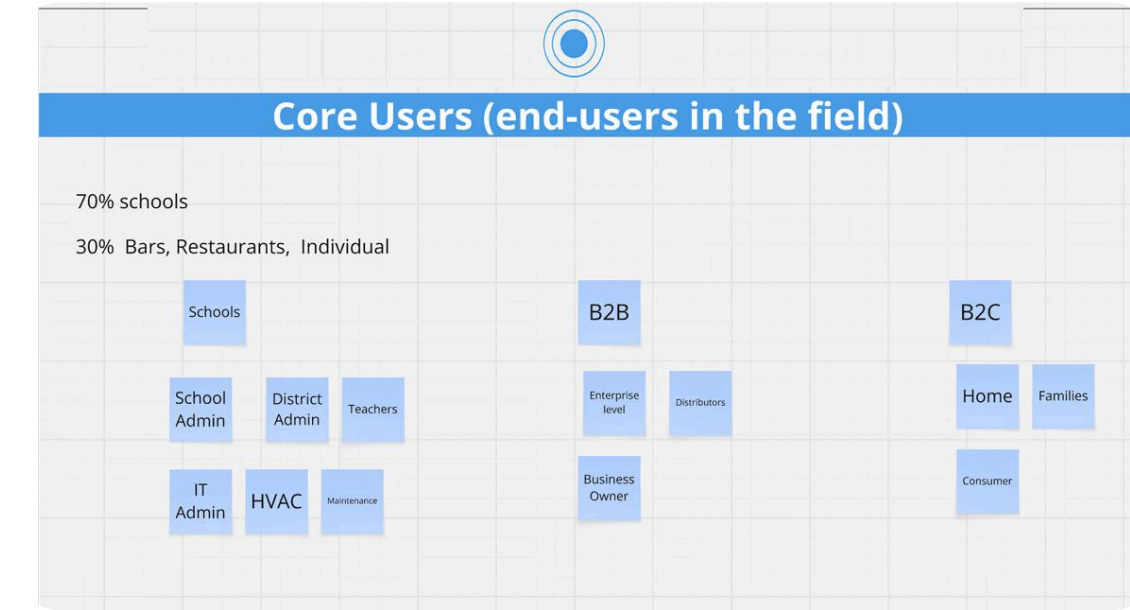
Competitive analysis examined existing air quality monitors like AirThings and iQAir as we studied their interfaces. This research help us understand effective data visualization patterns the right balance between data-intensive complexity and an intuitive user interface.

Technical domain research involved close collaboration with offering management, firmware teams, and engineers to understand sensor accuracy, appropriate alarm thresholds, s, and WiFi network integration constraints. We also considered numerous edge cases including offline scenarios, battery life considerations, and public WiFi access in corporate environments.

Accessibility evaluation contemplated visual alerts using traffic light colors on the physical device to ensure utilization of redundant cues like icons, text labels, and audio notifications with consistency across cross desktop and mobile applications. This research allowed us to accommodate for color-blind and low-vision users.

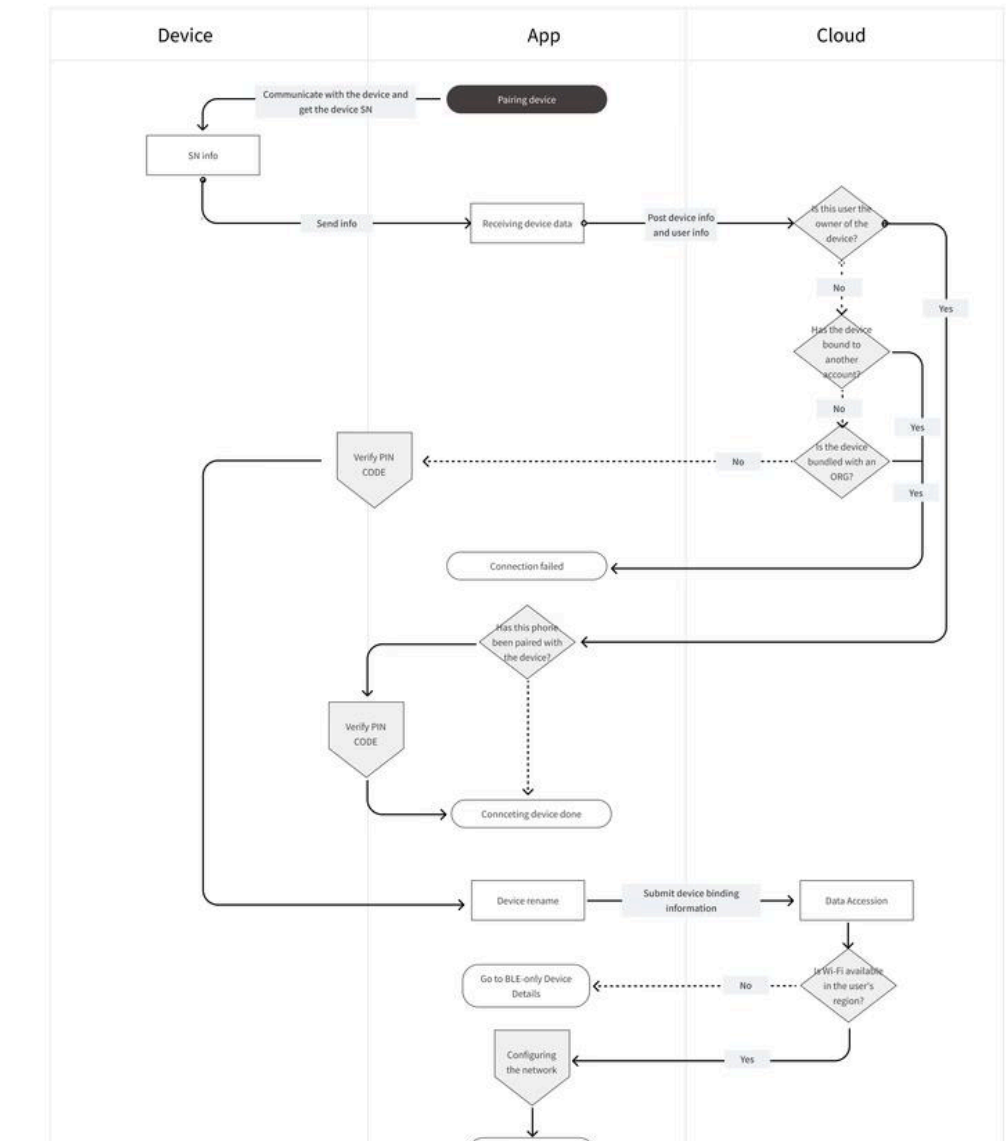


Product	Onboarding Experience	User Interface Design	Feedback & Alerts	Overall UX Takeaway
Airthings View Plus	Requires app pairing and dashboard registration	Clean but data-dense dashboard	Visual-only alerts	Excellent analytics, limited immediacy
uHoo Aura	Guided app setup, several permissions	Modern, data-rich interface	App notifications, no on-device cues	Great insights, but complex for schools

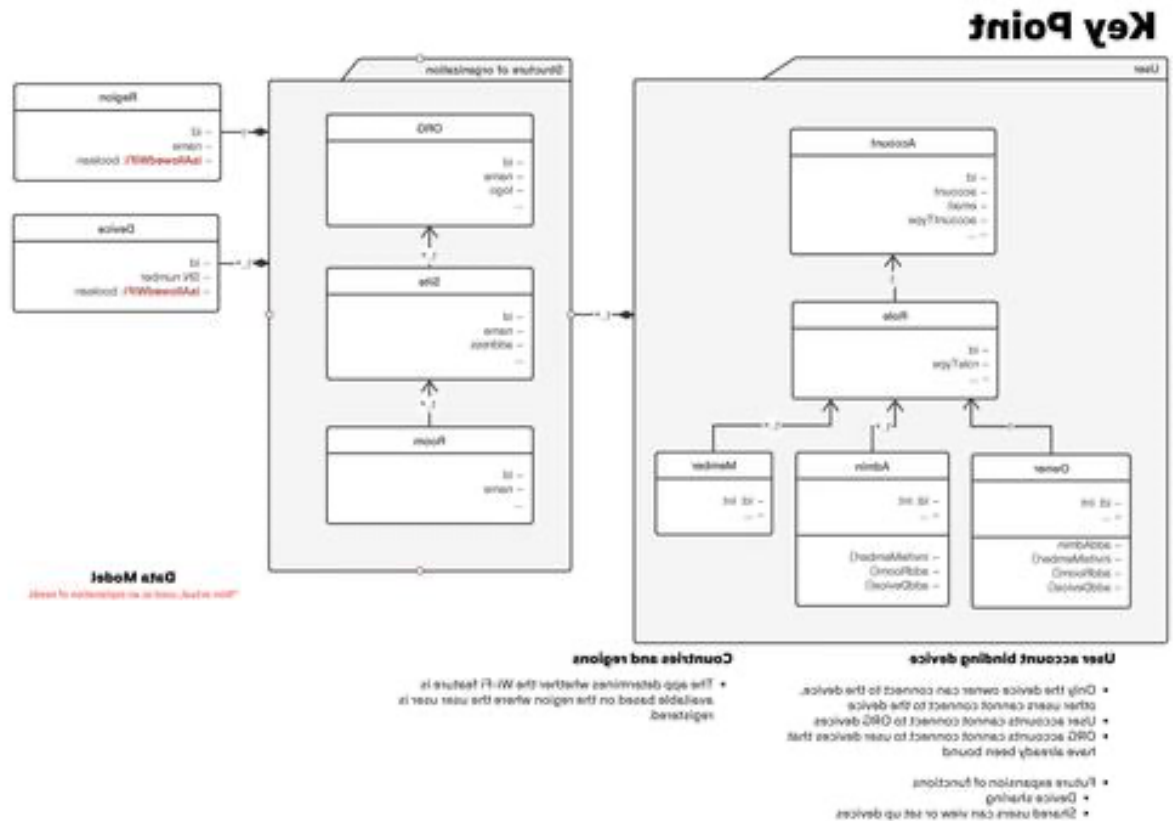
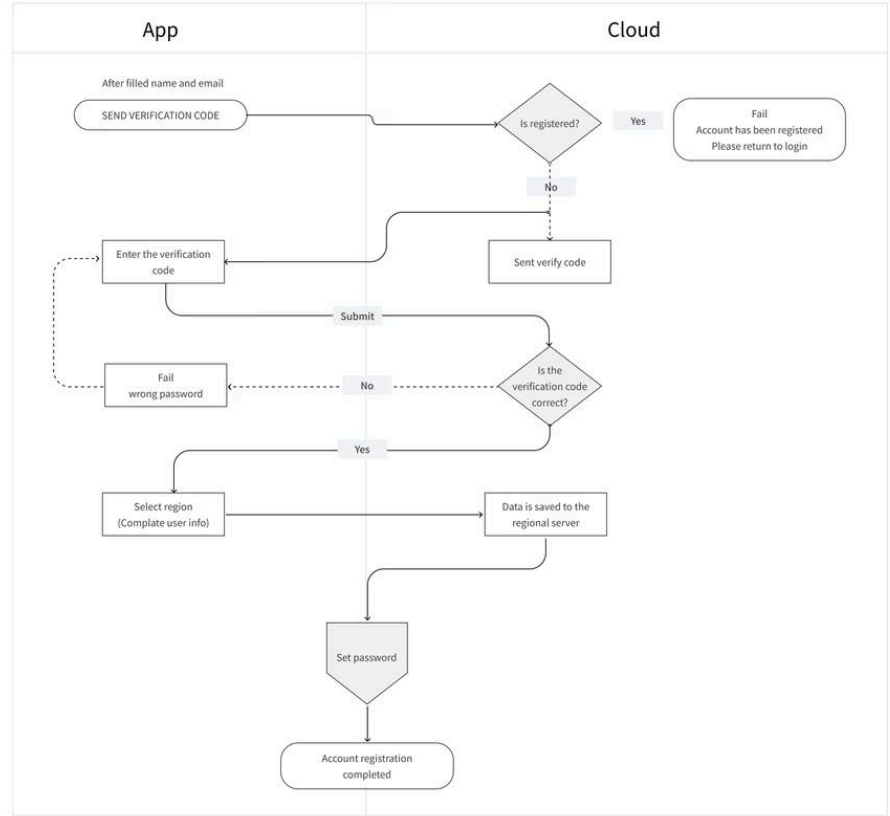


Workflows and Process Maps

During this phase, we worked through key workflow complexity by mapping Bluetooth pairing of brand new devices without existing owners and handling device access requests from both authorized and unauthorized users. We mapped out the self-registration process for Honeywell admins to onboard schools into the system. Additionally, we established sharing mechanisms that allowed device owners to grant access to other users. This process aided seamless device adoption across different user segments within school environments.



User Self-register



Outcomes and Impact

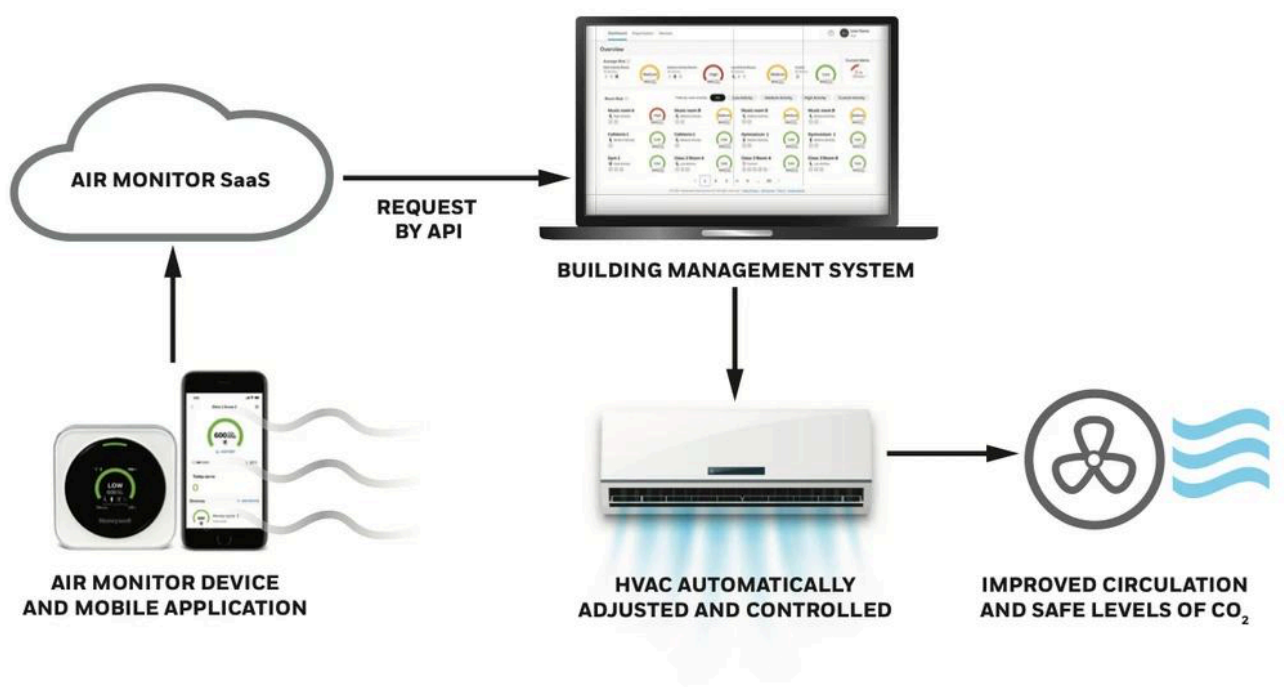
The device shipped with a simple, widely adopted UI that non-expert users could understand and act upon.

Positive outcomes from my work included:

- Increases in school administrator trust in indoor air quality monitoring in school / public settings.
- Facility team empowerment to setup devices seamlessly with existing infrastructure.
- Proactive teacher management of airborne illness risk in the classroom without disturbing the classroom.
- Air monitoring accuracy for school administrators for a holistic depiction of transmission risk at the school and event district level.
- Brand reinforcement for broader “Healthy Buildings” messaging within Honeywell.

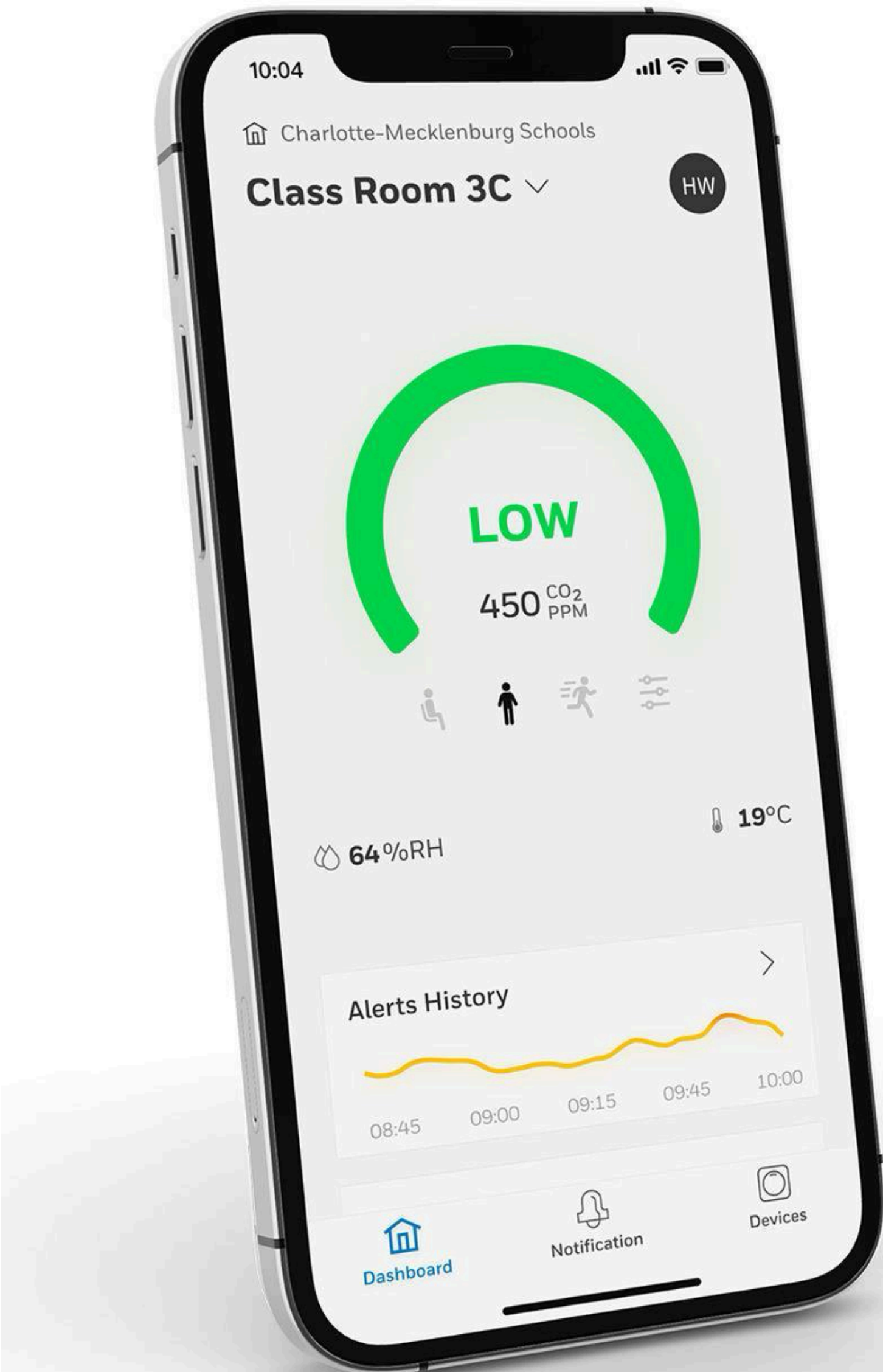
The project established foundational UX patterns for Honeywell air monitoring devices in educational environments, and the TRAM product was adopted in pilot school / building settings within Honeywell.

Bonus: The system’s integration with Honeywell’s BMS (e.g. Niagara) allowed automated ventilation adjustments – turning red alerts into active mitigation (this is documented publicly in a press release for the Charlotte Headquarters)



Reflection & Learnings

Designing for non-technical audiences pushed me to simplify the ui and setup process. I learned the importance of consistent and redundant signaling (beeps, icons, text) in real-world alert systems both on device and on-app. Balancing data richness vs usability taught me that I can’t expose too much sensor detail to overwhelm novice users. Collaboration across hardware, firmware, analytics, and building systems pushed me as a UX lead to be domain fluent This experience reinforced designing for trust, clarity, and inclusive communication in safety- and health-software solutions with IoT devices.

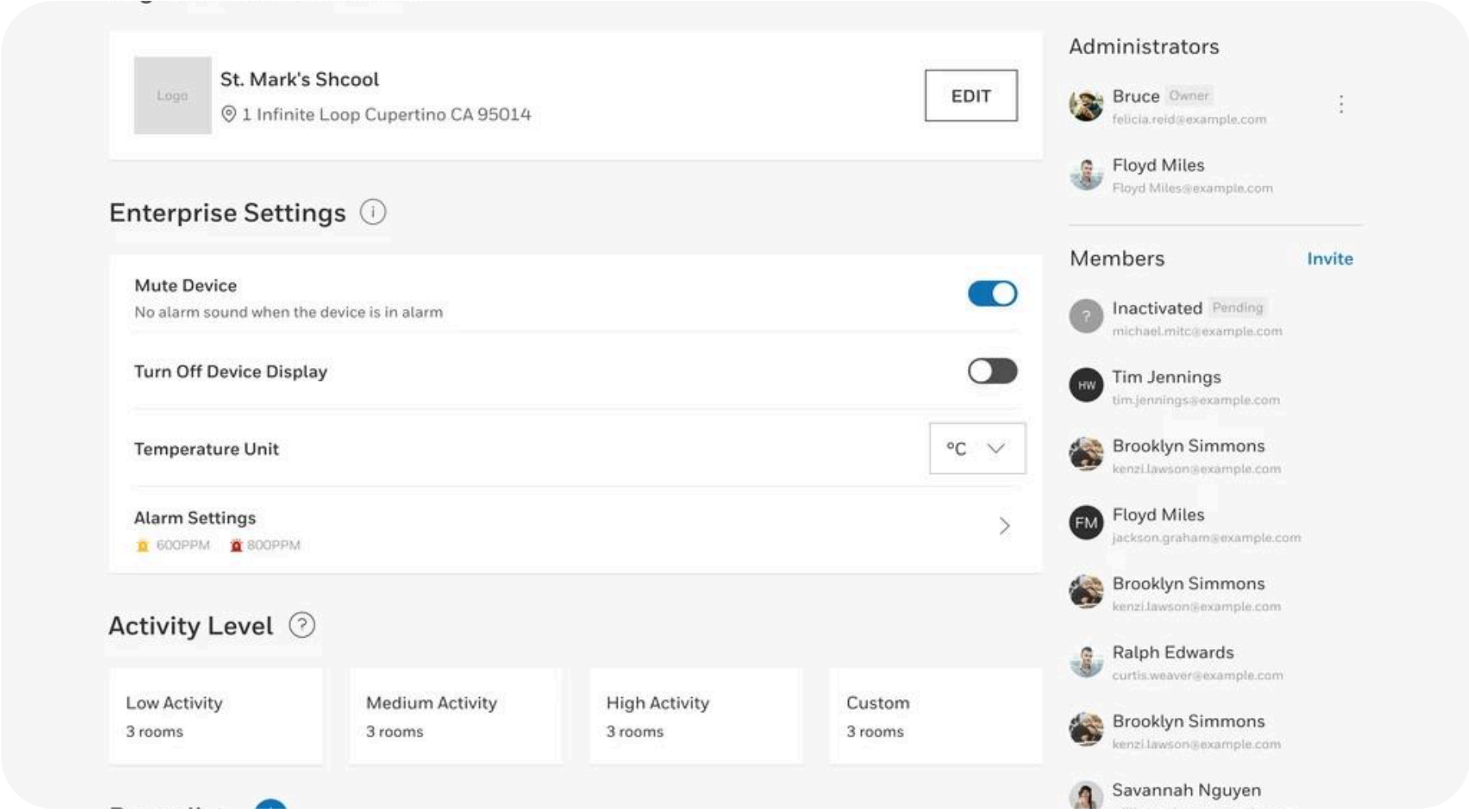
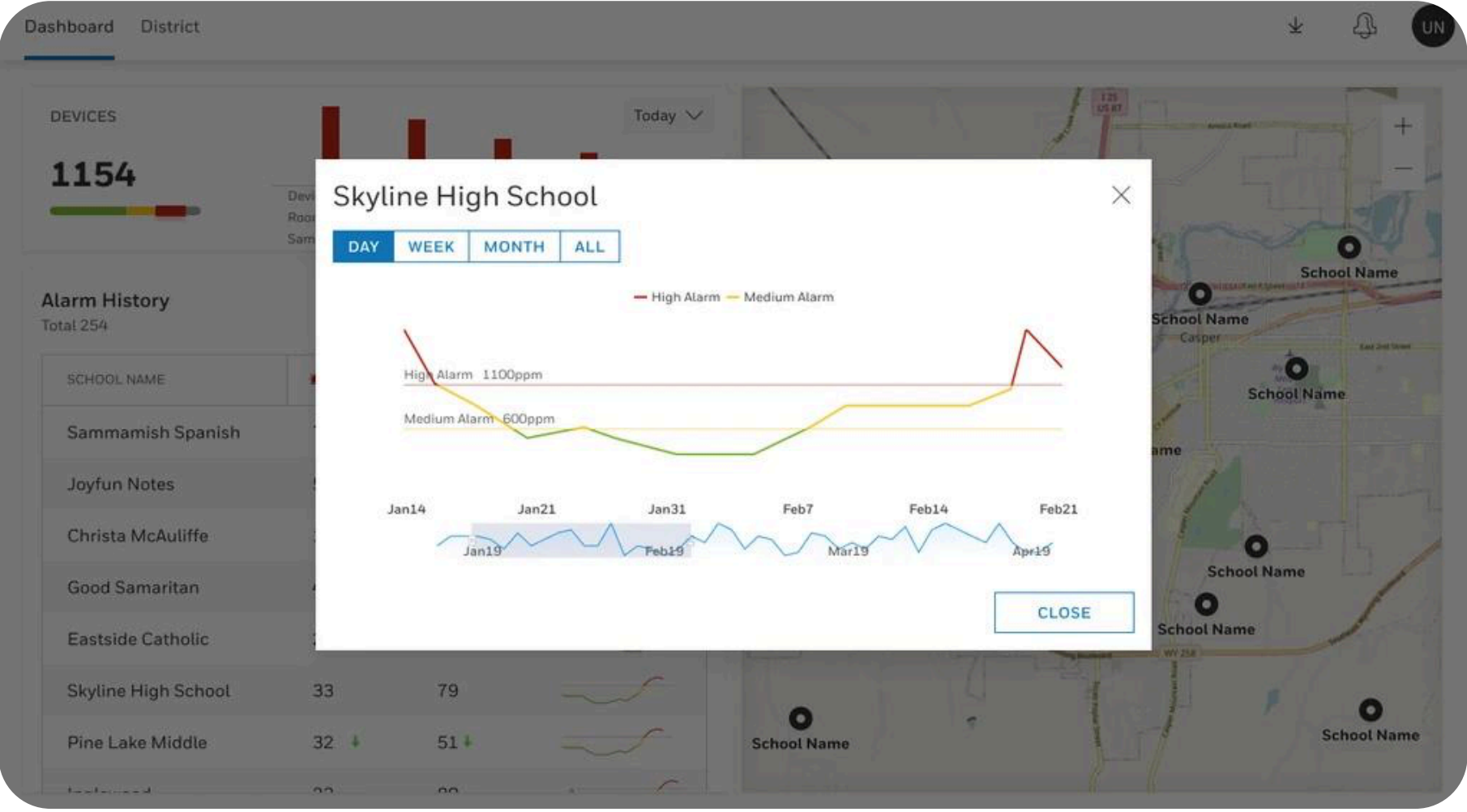
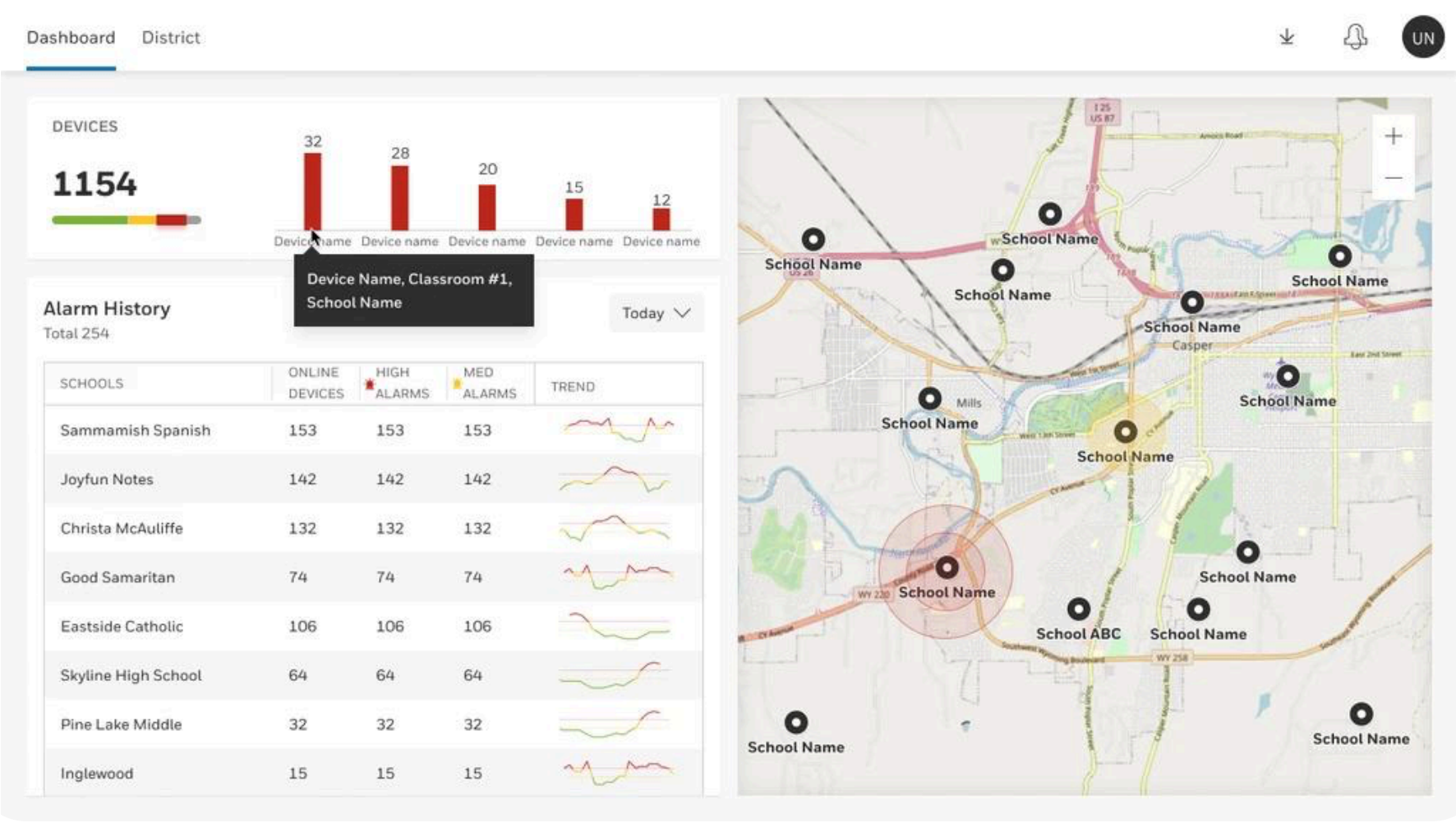
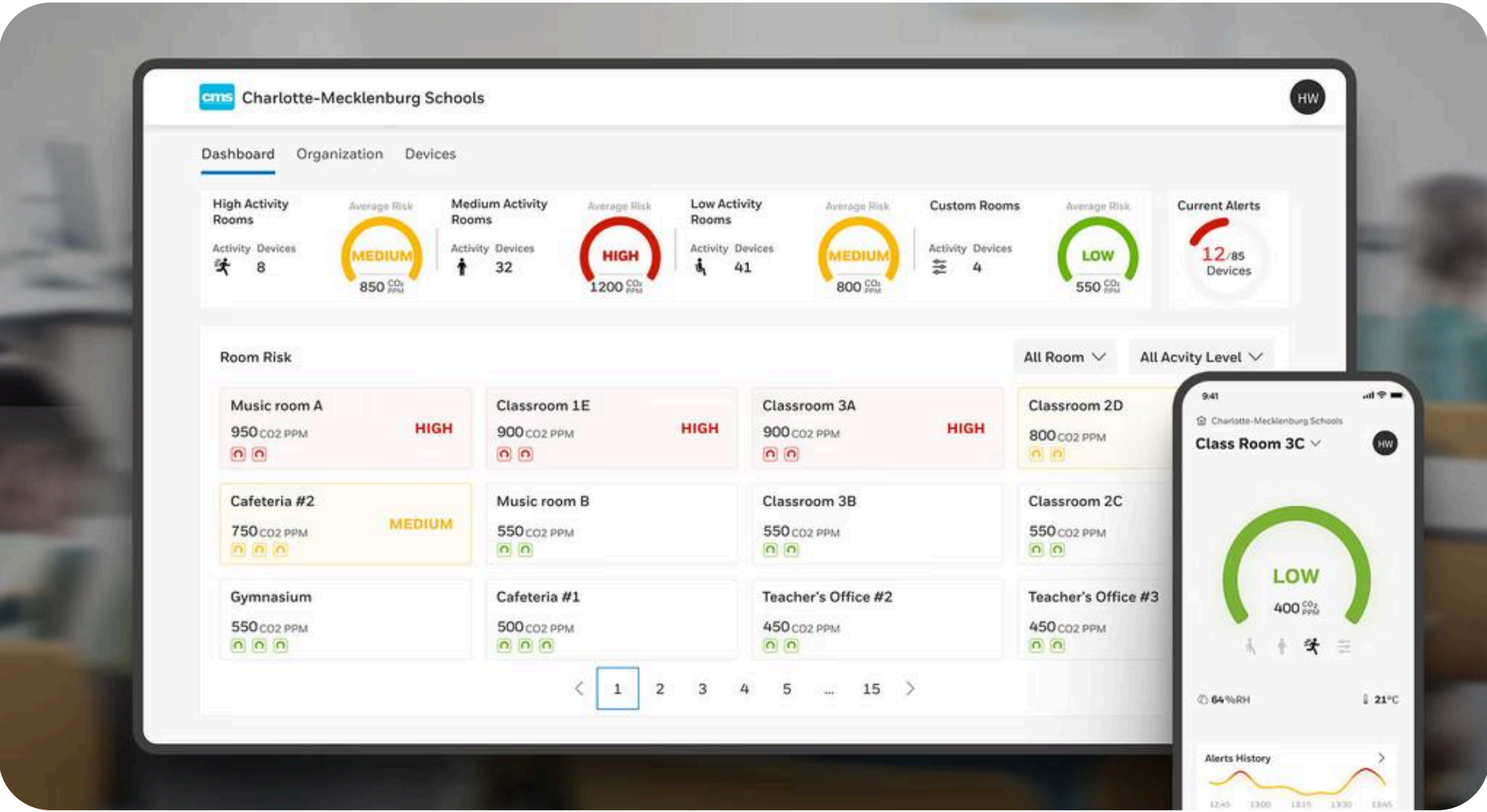


Validating and Testing Designs

I conducted usability testing with stakeholders, facility managers and education users using device prototypes and software. Highlights of this stage included:

- **Alert comprehension** — Testing whether users understood red/yellow/green indicators and knew what actions to take at each level
- **Design iteration** — Refining iconography, terminology, and data visualizations based on user feedback within hardware constraints
- **Integration testing** — Simulating alarm scenarios to validate dashboard responses and user workflows
- **Edge-case validation** — Testing battery warnings, network dropouts, and Bluetooth conflicts
- **Accessibility verification** — Checking contrast ratios and audible versus silent notification modes

Testing and validated informed design decisions and ensured an inclusive experience with insights clearly communicated to non-technical users.



Design Delivery

I help plan MVO1 as I led the design of comprehensive solutions across device, mobile, and web platforms throughout sprints 1-4. We delivered:

- Alert state systems** — Consistent visual language for green/yellow/red states across device and web platforms
- Onboarding flows** — Step-by-step user journeys for setting activity modes, linking devices to networks, and assigning room names and activity types
- Multi-device dashboards** — Comprehensive monitoring interfaces with trend graphs, alarm logs, and filtering capabilities across classrooms, schools, and device types
- Interactive prototypes** — Testable flows for device configuration, alert interpretation, and response actions used in usability validation sessions

These deliverables established a cohesive design system that scaled from individual device interactions to district-wide monitoring while maintaining consistency and usability for non-technical use

